DATA COLLECTION SYSTEMS

CANADIAN LIDAR

CloudPro

Point cloud georeferencing, cleaning, classification and delivery

Stage 03 of the Data Collection Systems chain

User Guide

for CloudPro 1.2.3

November 3, 2026

CloudPro — User Guide

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About CloudPro

CloudPro 1.2.3 is stage 03 of the Canadian Lidar Data Collection Systems (DCS) chain. It turns the mobile mapper's .clraw captures into a clean, classified, georeferenced point cloud for the job.

```
JobFolder/  (mapper captures, never changed)
  Processing/<Name>/<Name>.dcsproj      one project per job, shared by the
plugins
     01_PosPro/out/trajectory.json   ──┐   trajectory (SBET + SMRMSG at the
Certus reference point)
     02_AlignPro/out/alignment.json  ──┤   calibrated mounting, Lidar lever
arm, latency
     03_CloudPro/                      ▼
        cloudpro.json  filters/  work/ (point store, rebuildable)  log/
        out/  <Name>.las  <Name>_ground.tif  cloud_report.txt  cloud.json
```

What it does:

- **Georeferences** every Lidar return at its own firing time, with the trajectory PosPro published and the mounting, lever arm and latency AlignPro published.
- **Cleans and classifies** with fast manual tools in Plan, Slice and 3D views, and 36 automatic filters in groups with live preview.
- **Extracts road features**: markings, curbs, poles, traffic signs, power lines, manhole covers and guardrails.
- **Keeps up with the chain**: when PosPro or AlignPro publish again, *Update* rebuilds the points and carries every edit over.
- **Publishes** LAS 1.4, a ground grid and `cloud.json` for the job.

The raw captures are never changed. Menu: **Plugins ▸ Canadian Lidar ▸ Data Collection Systems ▸ CloudPro**.

Install

Install DCS Core first, then the stage plugins, then restart QGIS once.

1. In QGIS: **Plugins ▸ Manage and Install Plugins ▸ Install from ZIP**.
2. Install `dcs_core-<version>.zip`, then `pospro-<version>.zip`, `alignpro-<version>.zip` and `cloudpro-1.2.3.zip`, one zip at a time.
3. Restart QGIS. All three plugins share one menu: **Plugins ▸ Canadian Lidar ▸ Data Collection Systems**.

Requirements:

Item	Needed
QGIS	3.34 or later (QGIS 4 / Qt 6 supported)
Python packages	NumPy and SciPy (shipped with QGIS)
Reading .clraw	Zstandard is bundled for Windows Python 3.12 and 3.13; elsewhere the zstd library of QGIS/GDAL is used
Other plugins	DCS Core (required); PosPro and AlignPro for the full chain

Without DCS Core, CloudPro shows a message asking for it. This guide also ships inside the plugin: the **?** button, the project menu and **Help ▸ Plugins ▸ CloudPro user guide** open the copy that matches the installed version.

Quick start

A typical job takes five steps; PosPro (and ideally AlignPro) should have published first.

1. **Open the job.** Project button (centre of the top bar) ▸ *Open project…* and pick the job's `.dcsproj`, or click **03 CloudPro** in another plugin's chain bar. *New job project…* creates the shared project when CloudPro is the first plugin used.
2. **Georeference.** MENU ▸ *Georeference / update the points…* (Ctrl+I). Check the trajectory and mounting the dialog shows, then OK. The points are built in `03_CloudPro/work/`.
3. **Look around.** Drag the Inspection Area in the Plan view; the Slice and 3D views follow it.
4. **Clean and classify.** In the filters tab, pick *Mobile mapping – standard clean*, press **Live preview**, check the orange points, then **Accept** or **Run** on the whole project. Fix the rest by hand.

5. **Publish.** MENU ▸ *Publish for the job…* (Ctrl+P) writes the LAS, the ground grid and `cloud.json`.

When PosPro or AlignPro publish again, the chain bar shows CloudPro out of date and an **Update** button appears; press it, then publish again.

The window

The window has a top bar, a tool bar, four view quarters and a status strip.

```
 logo · MENU · 01 PosPro → 02 AlignPro → 03 CloudPro ....... PROJECT .......
inputs · Update · ?
 tools: hand · shape · line · ruler · i | action · inside/outside · class ·
line | undo · redo · rejected · zoom | CELL · POINT · BRUSH · COLOURS
 ┌──────────── PLAN ────────────┬──────────── 3D ────────────┐
 | project grid, drive lines,   | navigation arrows, orbit,  |
 | Inspection Area              | eye-dome lighting          |
 ├──────────── SLICE ───────────┼── AUTOMATIC FILTERS · LINES ─┤
 | cross-section or profile,    |   CLASSES · INFO · HISTORY |
 | slice bar on top             |                            |
 └──────────────────────────────┴────────────────────────────┘
 status: messages · cursor coordinates · point counts · progress · Cancel
```

Top bar

- **MENU**: Project properties (Ctrl+R), Georeference / update (Ctrl+I), Publish (Ctrl+P), Export (Ctrl+E), Save (Ctrl+S), pick the Inspection Area on the QGIS map, add the drive lines and Inspection Area to QGIS, add other .clraw files, memory limit, remove all points, keys and mouse (F1).
- **Chain bar** (01 / 02 / 03): the state of each stage of the job; click one to open that plugin on the same project.
- **Project button** (centre): new, open, recent, project properties, open the folder, relocate a moved job folder, save the QGIS project in the project folder, and *User guide (this version)*.
- **Inputs** line: which trajectory and calibration the points use, in amber when they are not the best available or out of date. **Update** appears only when the points are out of date.
- **?** (far right): opens this guide.

Tool bar. Drag its grip to move it or float it. Hover any icon for its name and key. The cycling buttons (selection shape, above/below line, action, inside/outside, rejected) change on each click; the small arrow in their bottom-right corner, or a right click, opens their list.

Splitters between the quarters can be dragged to resize the views.

Project properties

MENU ▸ *Project properties* (Ctrl+R) opens a window that can stay open or be minimised while you work. Nothing in it changes the raw captures.

Tab	What it shows	What you can change
Project information	Files and folders, created / saved, survey start and end, time recording, lines, survey length, points, extent, centre in latitude / longitude	–
Geodetics	Horizontal CRS, trajectory frame and epoch, ellipsoidal or orthometric heights	CRS (*Select…*), heights and geoid grid; optionally set the job's shared CRS. *Apply and re-georeference* rebuilds the points and keeps every edit
Survey configuration	Hardware recorded in the captures (Lidar, Certus, connections, lever arm, mounting), PosPro's trajectory, AlignPro's calibration	Source of the values used: chain default (AlignPro if published, else as recorded), as recorded, or entered (mounting, Lidar lever arm, latency)
Project files	One row per .clraw: line, start, duration, size	**Points** and **Track** tick boxes show or hide a file's points and drive line; clicking a row selects that file's points

Entered values are kept in `cloudpro.json` and marked in every output; results made without AlignPro say *no calibration*.

Views

All three views show only the points inside the Inspection Area; the Plan's background grid is the only thing drawn from the whole project.

View	What it shows	Header controls
Plan (top left)	The project grid (mean, max or min height, span, hits, rejects, intensity, or none) with hill shading, the drive lines with direction arrows, the Inspection Area and its points on top	GRID attribute, COLOUR, scale bar, LINK
Slice (bottom left)	A thin slab of the Inspection Area seen from the side: across it (a cross-section) or along it (a profile)	COLOUR, scale bar, LINK; slice bar below (direction, rotate 90°, whole area, THICK, STEP, previous / next, position slider)
3D (top right)	The Inspection Area, or only the slice	what to show, top / along / across / oblique presets, eye-dome lighting, COLOUR, scale bar, LINK

3D navigation arrows (top left of the 3D view): click an arrow head to turn the cloud that way — blue up / down (into the page), red left / right, green clockwise / anticlockwise on the screen. Each click turns 10°; hold to keep turning. The presets undo any on-screen turn. Mouse: left drag orbits, Shift+left or middle drag pans, wheel zooms, Ctrl+wheel scales Z, double click sets a new centre.

Colours are set per view: Height, Intensity, Intensity (range normalised), Classification, Line, File, Return, Channel, Time, Range. Rejected points are red; in a preview, orange = will be rejected and green = re-accepted; selected points are white. COLOURS on the tool bar sets the ramp.

Scale bars. Each view has its own colour scale bar, switched by the bar icon in its header: a ramp with values for height (absolute), intensity, time (UTC), range and laser, or the classes / lines / files present with their counts, plus rejected, preview and selected rows.

LINK makes a view follow another view's colour scheme and scale (Own / = Plan / = Slice / = 3D). A view cannot follow one that follows it. Scales are computed over the whole Inspection Area, so linked views agree.

CELL on the tool bar sets the Plan grid's cell: 0.5, 1, 2, 2.5, 5, 10, 12.5 or 25 m.

Navigation

The Inspection Area is the box you work in; every move can be made by an exact distance.

Inspection Area in the Plan (Pan tool, the hand): drag its edge to move it, a corner to resize it, the round handle to rotate it; Shift+drag draws a new one along the drag. Right click in the Plan:

- *Place the Inspection Area here*
- *Inspection Area properties…*: exact centre E/N, length, width and bearing; step forward / back / left / right by a set distance or by the length minus an overlap %; rotate by set degrees; follow a drive line (the area steps along its chainage and turns with it, and the chainage is shown); go to E/N; arrow / fly step for the views
- *Place the slice here*, *Align the Inspection Area with the nearest line*, *Zoom to the Inspection Area*, *Zoom to the whole project*

Slice. PgUp / PgDn or] / [step it by STEP (Shift = 5 steps). When a cross-section walks off the end of the area, the area moves on with it, so a corridor can be walked one slice at a time.

Other moves. Shift+arrows or Shift+PgUp / PgDn step the Inspection Area. Arrow keys pan a 2-D view by the set step; the wheel zooms at the cursor; Shift+wheel sets vertical exaggeration in the Slice. In 3D, W/S/A/D/Q/E fly by the step; 1 top, 2 along, 3 across, 4 oblique, O orthographic, 0 fit. MENU ▸ *Pick the Inspection Area on the QGIS map* centres it where you click the map.

Manual editing

Pick a tool, pick what it does, then draw in the Plan or Slice view (or right-drag in 3D).

Button	Options (key)
Hand	Pan (Space)
Selection shape	Rectangle (G), Polygon (P: click corners, right click or Enter to finish), Lasso (L), Brush (B)
Line	Above a drawn line (U), Below a drawn line (J)

Button	Options (key)
Ruler	Measure (M): distance, bearing, dH, dV, slope
i	Point information (I)
Action	Reject (X), Accept (V), Select (S), Classify (K, with the class beside it), Filter (F: runs the current filter group on the shape)
Inside / outside	Edit the points inside the shape, or the visible points outside it
Line list	Edit all lines, or only one

Selection. With Select, Shift adds to the selection. Right click a view for *reject / accept / classify / run the filter group on the selection*; Delete rejects it, Ctrl+D clears it.

Undo. Ctrl+Z / Ctrl+Y, without a fixed limit (up to 60 M changed points are kept). Every edit is written to disk at once and logged in the History tab. H shows or hides rejected points.

Rejected points are never deleted: they keep their reason (manual, automatic, original), so *Reset modifications* can re-accept, say, only the automatic ones.

Automatic filters

Filters live in groups that run in order, each on the result of the one before; nothing is committed until you accept or run.

Using a group. In the AUTOMATIC FILTERS tab, pick a group from GROUP (its **?** explains it; ☰ adds, duplicates, renames, imports, exports or deletes groups, and holds *Tune the filters…*). Pick a filter from FILTER: its settings fill the panel and its **?** gives what it does, every setting and the source paper. The switch turns a filter off; + − copy and the arrows add, remove, copy and reorder.

- **Live preview** recomputes in the background whenever a setting changes; **Pause** stops it; **Accept** commits it (undoable).
- **Run** applies the group to the SCOPE: Inspection Area, Slice, Selected lines (whole project) or Entire project. Whole-project runs go tile by tile with a buffer, so neighbourhood filters see across tile edges.

- Each filter can be limited to some classes and lines. Mode *Individually* filters each line on its own.

Ready-made groups: Mobile mapping – standard clean · Snow and rain · Ground and vegetation classes · Moving objects (multi-pass) · Feature extraction (road and street furniture) · Road surface QA. Groups are saved in `filters/groups.json` and can be shared as JSON.

Kind	Filters
Edit	Reset modifications, Edit classification, Shift height
Clip	Absolute height, intensity (absolute / relative), range and sensor geometry, laser channel / return, time, reference surface (a QGIS raster), vehicle self-hits
Cell statistics	95 % confidence level, low hit count, mean height, lonely cells
Reduce	Voxel reducer, scan-based reducer
Outliers	Statistical outlier (SOR), radius outlier, snow / rain / dust (DROR, LIOR, DIOR), isolated cluster, slope / spike, surface fit
Structure	Vertical plane (walls, reflections behind glass), geometric features (eigenvalues)
Ground	Inverted cloth (CSF), SMRF, PMF, height above ground (low noise, vegetation 3/4/5, bands)
Multi-pass	Moving objects / ghosts
Features	Road markings (68), curbs (69), poles (65), traffic signs (67), power lines (14), manhole covers (70), guardrails (71)

Feature extraction classifies by default and needs ground first (the ready-made group starts with SMRF). Intensity thresholds were set on simulated XT-32 data; check them on your first real job.

Tune to the sensor precision. ☰ ▸ *Tune the filters to the sensor precision and this data…* measures the noise on the ground of the loaded Inspection Area and combines it with the hardware error budget (XT32 range precision 0.5 cm, Certus Evo attitude and RTK position, or PosPro's own standard deviations). It lists every setting that would change, old → new and why, and applies to this group or all groups. Height and distance tolerances follow the noise; feature settings that describe the objects are only ever raised. Run it on a road seen by two or more passes.

Georeferencing and updates

CloudPro reads only what PosPro and AlignPro published for the job, and records which versions it used.

Input	Used	When it is missing
Trajectory	PosPro's `trajectory.json` (SBET + SMRMSG, frame and epoch)	The Certus real-time track, with a warning (metres of drift without RTK)
Mounting, Lidar lever arm, latency	AlignPro's `alignment.json`	The mounting recorded in the capture; every result says **no calibration**
Captures	The job's .clraw files, found from the shared project	MENU ▸ *Add other .clraw files…*

The Georeference dialog also sets the returns, range gate, thinning and an optional geoid grid.

Out of date. When PosPro or AlignPro publish again, the chain bar shows 03 CloudPro out of date, the inputs line says why, and **Update** appears. Update rebuilds the points with the new inputs and re-applies every manual and filter edit, matched by firing time, laser and return.

Moved data. If the captures moved (a new drive letter, a copied folder), CloudPro asks for the job folder once; the project button also has *Raw data moved? Relocate the job folder…*.

Rules. The raw captures are never changed. `work/` can be deleted at any time; Georeference rebuilds it. While a job's cloud is open for editing, a second QGIS refuses to edit it. Each run writes a log in `log/`.

Publish and export

Publish is the job's delivery; Export writes a copy anywhere.

Publish (MENU ▸ *Publish for the job…*, Ctrl+P) writes to `03_CloudPro/out/`, all or nothing:

- `<Name>.las` (LAS 1.4) — with or without rejected points, all or chosen classes;
- `<Name>_ground.tif` — optional ground grid at a chosen cell;
- `cloud_report.txt` and, last, `cloud.json`, which records the trajectory and alignment the points were really georeferenced with, *no calibration* when AlignPro was not used, the filter groups applied with their settings, point and class counts, the files, CRS and heights.

Publishing after PosPro or AlignPro changed, without updating, still shows CloudPro out of date. *Add to QGIS* loads the outputs into the map.

Export (MENU ▸ *Export…*, Ctrl+E): LAS, XYZ or a grid (GeoTIFF) of the Inspection Area or the whole project, chosen lines and classes, one file per line if wanted. In the LAS, the laser id goes in *user data* and the line in *point source id*.

Classes follow ASPRS LAS 1.4, plus 64 Vehicle (dynamic), 65 Pole, 66 Snow / rain, 67 Traffic sign, 68 Road marking, 69 Curb, 70 Manhole cover, 71 Guardrail.

Large jobs, memory and stability

CloudPro keeps big jobs responsive by drawing only what the computer can draw smoothly and by checking size before loading.

- **Point budgets.** While you pan, zoom or orbit, each view draws only as many points as keep it near 30 frames per second, measured from its own frame times. A still 3D frame is capped at about 0.3 s and shows "showing N %". Points are loaded in random order, so the subset is an even sample. Picking and selecting always use every point.
- **Memory check.** Before an Inspection Area is loaded, its size is estimated from the project grids without reading any points. Above the limit (15 million points by default; MENU ▸ *Memory limit*) CloudPro asks first.
- **Tiles.** The point store is 50 m tiles; only tiles touching the Inspection Area are opened, and at most 256 stay open.
- **Autosave** every 2 minutes; filter groups are written to a temporary file and renamed, so a crash cannot leave a half-written file.

- **Whole-project filter runs** go tile by tile and hold the stage lock; very large runs cannot be undone, but *Reset modifications* reverts automatic rejections.

Batch processing

The same steps run without the window, from the OSGeo4W shell in the QGIS plugins folder (which holds `dcs_core` and `cloudpro`):

```
python -m cloudpro.cli import  D:/Job/Processing/Street/Street.dcsproj
(PosPro + AlignPro inputs)
python -m cloudpro.cli update  D:/Job/Processing/Street/Street.dcsproj
(re-georeference, keep edits)
python -m cloudpro.cli filter  D:/Job/Processing/Street/Street.dcsproj --
group "Mobile mapping - standard clean"
python -m cloudpro.cli publish D:/Job/Processing/Street/Street.dcsproj --dem
0.5
python -m cloudpro.cli info    D:/Job/Processing/Street/Street.dcsproj
(chain state + what is out of date)
```

The command line uses the same project, inputs, entered values and filter groups as the window.

Keys and mouse

Key / mouse	Does
Space, G, P, L, B	Pan, rectangle, polygon, lasso, brush
U, J	Above line, below line
M, I	Measure, point information
X, V, S, K, F	Reject, accept, select, classify, filter
H	Show / hide rejected points
Delete, Ctrl+D	Reject the selection, clear the selection
Ctrl+Z, Ctrl+Y	Undo, redo
PgUp / PgDn,] / [	Next / previous slice (Shift = 5 steps)
Shift+arrows, Shift+PgUp / PgDn	Step the Inspection Area

Key / mouse	Does
Arrows	Pan a 2-D view by the step
Wheel, Shift+wheel	Zoom at the cursor, vertical exaggeration
Middle drag, double click, Home	Pan in any tool, centre here, zoom to the points
Ctrl+Home	Zoom all views to the Inspection Area
3D: left drag, Shift / middle drag, right drag	Orbit, pan, lasso or rectangle edit
3D: W/S/A/D/Q/E, 1 / 2 / 3 / 4, O, 0	Fly, top / along / across / oblique, orthographic, fit
Ctrl+R, Ctrl+I, Ctrl+P, Ctrl+E, Ctrl+S	Properties, georeference, publish, export, save
F1	Keys and mouse

Troubleshooting and limits

Symptom	What to do
"CloudPro needs DCS Core"	Install DCS Core, restart QGIS
No .clraw files found	Check the job folder in the project; MENU ▸ *Add other .clraw files…* or *Relocate the job folder…*
Inputs line in amber, "no calibration"	AlignPro has not published; run it, then **Update**
"This job's point cloud is open elsewhere"	Close it in the other QGIS (one editor per job)
Inspection Area not loaded	It is over the memory limit: make it smaller or raise MENU ▸ *Memory limit*
3D shows "showing N %"	Normal on big areas; picking and editing still use every point
Features missed or over-found	Run *Tune the filters…*, then adjust the intensity thresholds in the filter's settings

Limits to know

- Tested on simulated captures and on QGIS 3.34 under Linux; check the first real job against another processing of the same capture.
- The 3D view is software rendered (no graphics drivers needed) and stays fluid up to about 2–3 M points on a typical laptop.
- The LAS scan-angle field is 0.
- DROR's angle must match the data's azimuth step: 0.18° for the XT-32 at 600 rpm, ×N when keeping every Nth packet.

References. Every filter's **?** names its source paper; the full list is in `docs/REFERENCES.md` in the plugin folder.